

# Multimodal Graph Conditioned Diffusion Model for Video Captioning

Benhui Zhang  
Northwestern Polytechnical  
University  
Xi'an, China  
zhangbh0208@gmail.com

Junyu Gao  
Northwestern Polytechnical  
University  
Xi'an, China  
gjy3035@gmail.com

Yuan Yuan\*

Northwestern Polytechnical  
University  
Xi'an, China  
y.yuan1.ieee@gmail.com

## Abstract

Video captioning aims to describe the content of a given video with condensed natural language sentences. Such a captioning task is full of challenges since the high requirements for visual-textual relevance and multimodal fusion understanding. Previous works primarily focus on visual content modeling, often overlooking the rich semantic correlations between visual and textual modalities, which results in incomplete understanding of the multimodal context and suboptimal caption accuracy. In this paper, we propose a multimodal graph conditioned diffusion model for video captioning, named *MGCDVc*. The idea behind our model is to incorporate graph-based relational reasoning with diffusion-based generative modeling to jointly model cross-modal relationships and capture latent semantic structure. Specifically, we learn a set of latent concept anchors to bridge the visual and textual modality nodes, enabling the construction of a weighted multimodal graph. Then we introduce the graph conditioned diffusion strategy which generates the textual semantic nodes and associated edges under the graph structure awareness condition. Furthermore, a soft pruning mechanism is designed to filter out low-quality nodes, thus further refining the generated multimodal graph to provide more accurate semantic structural guidance for caption generation. Experimental results on several popular datasets demonstrate that our model achieves better performance in video captioning task.

## CCS Concepts

• Computing methodologies → Natural language generation; Computer vision representations; • Theory of computation → Semantics and reasoning.

## Keywords

Multimodal Learning, Language Generation, Graph Neural Network

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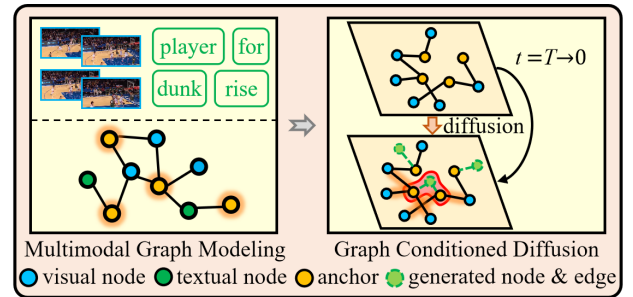
\*Corresponding author



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## 1 Introduction

With the rapid growth of user-generated content on platforms such as YouTube, TikTok, and Instagram, understanding and describing video content at web scale has become increasingly important. Video captioning aims to generate descriptive natural language for video content by modeling the temporal and semantic dynamics of visual sequences. As a meaningful cross-modal task, it bridges visual and textual information and serves as a core component in many open-domain web scenarios, such as video retrieval, content recommendation, and human-AI interaction on social media platforms. However, the inherent temporal dynamics of videos and the semantic gap between visual features and textual expressions inevitably pose challenges in accurate caption generation. A recent line of approaches [41, 55, 62] have explored mining latent multimodal semantic features, aiming to achieve a better understanding of how visual signals align with natural language semantics. Although significant progress has been made, previous work may not focus sufficiently on the salient information within videos and fails to preserve the structured semantic of these important regions.



**Figure 1: Illustration of multimodal graph modeling and graph conditioned diffusion.** We first learn a set of latent semantic concept anchors to bridge the visual and textual modalities for multimodal graph construction. Then the textual semantic nodes and related edges are generated via the designed graph conditioned diffusion strategy.

Graph-based representation learning [22, 48] is well suited for capturing complex relationships and dependencies between objects, actions, and concepts in video and text [23, 42], enabling a better understanding of the intricate interactions and dynamics within multimodal content. Recently, graph-based approaches have shown promising results in video captioning task, since graph representation learning enables a better understanding of the intricate

interactions between visual and linguistic elements, providing a structured way to capture the complex relationship and dependency within multimodal content [23, 42]. [57] constructs spatio-temporal graphs to depict objects with their relations within the frames. [59] adopts an object relational graph-based encoder to capture more detailed interaction features for enriched visual representation. By constructing an object-action graph, [47] describes the shallow relationship between objects and actions in the video. However, previous graph-based video captioning works often focus solely on visual information, neglecting the interaction between visual and textual information, and they often suffer from fixed graph topology and limited relational modeling capabilities.

Moreover, the diffusion model [11] has emerged as a new generative framework capable of modeling data distribution through iterative denoising, which naturally enables dynamic structure refinement, achieving promising results in various generative tasks [7, 13, 35]. However, conventional diffusion models lack explicit mechanisms to capture the complex relational and topological structures inherent in graph-structured multimodal data, which are crucial for semantic understanding in cross-modal tasks. In response to this limitation, the graph diffusion model [9] extends the diffusion paradigm to graph-structured data, where the diffusion process is guided not only by the individual features of the nodes but also by the connectivity patterns encoded in the graph structure, enabling the model to capture both local interactions and global dependencies.

To capture the semantic correlation between video and text, in this paper, we introduce a multimodal graph conditioned diffusion model for video captioning that integrates the strengths of diffusion-based generative modeling with graph-based relational reasoning. As shown in Figure 1, we first construct a weighted multimodal graph for multimodal modeling based on a set of learned latent concept anchors, which serves as a medium bridge between visual and textual modalities to promote the cross-modal consistent representation learning. Then we introduce the designed multimodal graph conditioned diffusion strategy for visual counterparts learning with graph structure awareness condition, since the text is expected to be generated during inference based on the video contents, this strategy provides a relational semantic cue of the visual input for the downstream caption generation task. In addition, a soft pruning mechanism is designed to further improve the quality of the generated graph by filtering out low-validity nodes based on predicted validity scores. By reducing structural redundancy and highlighting meaningful semantics, this mechanism contributes to more contextually relevant caption generation. The final obtained multimodal graph is utilized for the following caption generation. In all, the main contributions of this study are summarized as follows:

- We propose a multimodal graph conditioned diffusion model for video captioning, where a weighted multimodal graph is constructed based on a set of learned latent concept anchors that serve as a shared medium bridging visual and textual modalities, enhancing alignment and benefiting downstream caption generation.
- We introduce a graph conditioned diffusion strategy to enable cross-modal semantic relation modeling and to enhance textual semantic node representation through structure-aware

denoising, thereby providing contextually grounded and structured semantic guidance for caption generation.

- A soft pruning mechanism is designed to refine the generated graph by filtering out low-quality nodes based on the valid score, thus reducing structural redundancy and promoting semantic focus in caption generation.
- Experiments on several datasets demonstrate the effectiveness of the proposed model, which can achieve better performance compared to the baselines.

## 2 Related Works

### 2.1 Video Captioning

Early video captioning research adopts the template-based strategy, which usually aligns the sentence components to the detected visual content, and obtains the description for the input video with the pre-defined templates [20, 50]. However, an obvious problem is that the generated sentences are limited by the fixed templates. Sequence learning-based methods, adopting an encoder-decoder structure, have gradually become a popular solution for video caption task. Since without bounded by the pre-defined template, such methods can generate captions with more flexible sentence patterns. Specifically, [4, 34, 54] are typical methods that utilize the LSTM/GRU structure to serve as a decoder to generate captions. [16] further enhance the current encoder-decoder framework by incorporating implicit external knowledge with neural memory. [15, 55] adopt a transformer-based network for video captioning. [53] propose a hierarchical framework to encode visual content and bridge visual-textual counterparts from multiple levels. [25] propose a long short-term relation transformer with global gating to model long-term temporal dependencies in the video. Different from the above methods, we conduct the diffusion process based on the constructed multimodal graph to capture cross-modal consistency and visual-text association relationship, thereby enhancing the quality of caption generation.

### 2.2 Multimodal Representation Learning

Learning effective cross-modal representation is one of the key challenges in multimodal learning. By modeling the interactions between different modalities more effectively, deep learning models can capture more comprehensive information. Extensive research has shown the promising potential of multimodal representation learning, particularly in vision language tasks [1, 55]. Masked language modeling has also been explored in multimodal contexts. Studies such as [21, 27, 51] adopt a unified transformer encoder and employ multitask learning to achieve effective alignment across modalities. Moreover, [34, 55, 62] pay more attention to multimodal semantic consistency learning. [34] further improve the semantic correlations for video and language representations to generate semantically more accurate captions. In this paper, we characterize multimodal information through graph structure, and gain a deeper understanding of the correlations across modalities through the diffusion process, in which the break and reconstruction of the edge between visual and textual nodes is guided based on the correspondence, thereby obtaining a better understanding of the mapping relationship between visual-text pairs.

### 2.3 Diffusion Model for Generation

Diffusion model has demonstrated promising performance in generation tasks [11, 33, 36]. Recent studies have explored the diffusion process in various domains. For instance, [12, 29] leverage diffusion model for molecular generation. In the field of recommendation, [14, 35] utilize diffusion-based approach to capture user-item interactions across multiple modalities. Furthermore, [7, 46] extend the diffusion model to the non-Euclidean manifold space, enabling a more effective modeling of complex data distributions. For the caption task, [33, 61] introduce the comprehensive semantic information of the input images into the diffusion process, which act as semantic prior to guide the learning of each reverse state transition. [28] inject prefix image embeddings into the denoising process  $\epsilon$  to generate more accurate captions. In this paper, we explore the diffusion model in multimodal graph structure to capture the mapping correlation between visual nodes and textual nodes, thus obtaining a better understanding of the correspondence between the two modalities and achieving better caption results.

### 3 Methodology

We propose a multimodal graph conditioned diffusion model for video captioning (*MGCDVc*), as shown in Figure 2, which consisted of four modules: (a) **Uni-modal Encoder** takes video and textual description as input and generates the corresponding visual and textual representations, respectively. (b) **Multimodal Graph Construction** utilizes the learned uni-modal representations as well as a set of extra learned latent semantic concept anchors to construct a weighted multimodal graph. (c) **Graph Conditioned Diffusion** exploit the diffusion process to learn the semantic counterpart of the visual content based on the constructed multimodal graph. (d) **Caption Generation** takes the learned features to predict the caption results.

#### 3.1 Uni-modal Encoder

Given the input video  $V$ , we follow the setting in [32] for the extraction of the video features. The input video is first sampled with  $T$  frames, each with a size of  $H \times W \times 3$  pixels. Then the sampled frames with a size of  $T \times H \times W \times 3$  are inputted to the pre-trained video transformer [32], which outputs the video feature  $V \in \mathbb{R}^{n_v \times d_v}$  serving as the global visual feature, where  $n_v$  and  $d_v$  denote the number of visual feature vectors and dimensions, respectively. For text embedding, the input text sentence is first tokenized into a sequence of tokens. Then a lightweight word embedding layer is applied to transform these tokens into the corresponding text representation  $T \in \mathbb{R}^{n_t \times d_t}$ , where  $n_t$  and  $d_t$  denote the number of textual feature vectors and dimensions, respectively.

#### 3.2 Multimodal Graph Construction

Since the visual and textual features are extracted separately by uni-modal encoders, directly utilizing these unaligned representations for the downstream caption generation task may lead to inaccurate results and degraded performance. Graph offers an effective way for cross-modal understanding by modeling intermodal relationships, enabling feature propagation between nodes to preserve unique characteristics while extracting shared patterns, thus improving robustness and performance. In this module, we construct a weighted

multimodal graph to explore the relationships between video content and textual semantic for effective multimodal representation learning. To further bridge the gap between the visual and textual modalities, we first learn a set of latent semantic concept anchors that explicitly establish associations between visual content and textual semantic, denoted as  $A = \{a_1, a_2, \dots, a_{n_a}\}$ , where  $A \in \mathbb{R}^{n_a \times d_a}$ ,  $n_a$  and  $d_a$  denote the number of anchors and dimensions, respectively, each anchor  $a_i$  captures a shared latent semantic concept across modalities. Then we construct the weighted multimodal graph  $G = (V, E, A)$ , where  $V = \{V^v, V^a, V^t\}$  is the set of three types of nodes,  $E = \{E^{va}, E^{ta}\}$  is the set of edges between these nodes, and  $A = \{A^{va}, A^{ta}\}$  is the set of the adjacency matrices. To quantify the relationships between the anchors and visual and textual semantic, we define the weighted adjacency matrix  $A^{va}$  and  $A^{ta}$ , which represents the relational weights between visual and textual vectors with anchors, establishing the correspondence between specific visual feature and corresponding latent semantic, thus improving the alignment and coherence between visual content and text. The element of  $A^{va}$  and  $A^{ta}$  are formed as:

$$A_{ij}^{va} = \frac{\exp(\phi(V_i)A_j^T)}{\sum_{j=1}^{N_a} \exp(\phi(V_i)A_j^T)},$$

$$A_{ij}^{ta} = \frac{\exp(\varphi(T_i)A_j^T)}{\sum_{j=1}^{N_a} \exp(\varphi(T_i)A_j^T)},$$
(1)

where  $\phi$  and  $\varphi$  are the trainable project matrices to project the video representation  $V$  and the text representation  $T$  into the same semantic space with anchors. Moreover, to promote the consistence and alignment between the visual content and textual semantic over the learned anchors, we set the alignment loss, formulated as:

$$\mathcal{L}_{ali} = \frac{1}{D} \sum_{i=1}^D (\mu_i^v - \mu_i^t)^2,$$
(2)

where  $\mu^v = \frac{1}{n_v} \sum_j A_j^{va}$  and  $\mu^t = \frac{1}{n_t} \sum_j A_j^{ta}$  denote the average attention distributions of visual and textual features over the shared anchors, respectively. By aligning their focus on the same set of anchors, this loss facilitates the construction of a unified semantic space across modalities, which serves as the foundation for the downstream caption task.

#### 3.3 Graph Conditioned Diffusion

To integrate the strengths of diffusion-based generative modeling with graph-based relational reasoning, we introduce a multimodal graph conditioned diffusion module that generates semantically aligned textual representation based on cross-modal interaction, where the multimodal graph information is incorporated as conditional input to the diffusion process, guiding the generation of textual representations (i.e., visual counterparts) based on the visual input and intermediate anchors.

**3.3.1 Forward Process.** As captions are not available and need to be generated during inference, we adopt a denoising approach by progressively adding noise to the text nodes and their corresponding edges during training, encouraging the model to reconstruct the associated visual counterparts for the following caption generation task. In specific, we take target nodes and edges as the basic variables for generation and perform forward diffusion on the node

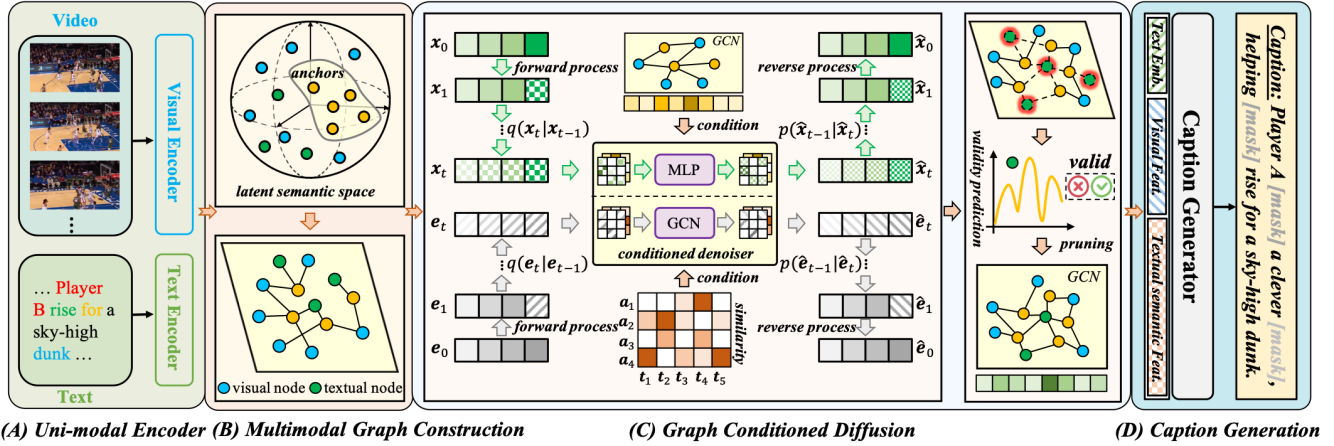


Figure 2: The overall architecture of the proposed model. We first extract uni-modal features, construct a weighted multimodal graph with a set of learned latent concept anchors, and obtain the visual semantic guided cues via the designed graph conditioned diffusion strategy for final caption generation.

features and edge weights of the graph at the time step  $t$ . We initialize the forward diffusion process with  $x_0 = \varphi(T)$  and  $e_0 = A^{ta}$ . The forward diffusion process constructs  $x_{1:T}$  and  $e_{1:T}$  in a Markov chain by progressively adding Gaussian noise over  $T$  steps. The transition from  $x_{t-1}$  to  $x_t$  and  $e_{t-1}$  to  $e_t$  can be formulated as:

$$\begin{cases} q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t\mathbf{I}) \\ q(e_t|e_{t-1}) = \mathcal{N}(e_t; \sqrt{1 - \beta_t}e_{t-1}, \beta_t\mathbf{I}) \end{cases} \quad (3)$$

The forward diffusion process consisted of  $t$  diffusion steps, where  $t \in 1, 2, \dots, T$ ,  $\mathcal{N}$  denotes Gaussian distribution, and the scale of Gaussian noise added at each step is controlled by  $\beta_t \in (0, 1)$ . With the reparameterization trick, we can directly obtain the final step  $x_t$  from  $x_0$  by aggregating the noising process at each step. Then the forward process can be reformulated as:

$$\begin{cases} q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)\mathbf{I}) \\ q(e_t|e_0) = \mathcal{N}(e_t; \sqrt{\bar{\alpha}_t}e_0, (1 - \bar{\alpha}_t)\mathbf{I}) \end{cases} \quad (4)$$

where  $\bar{\alpha}_t = 1 - \beta_t$ , and  $\bar{\alpha}_t = \prod_{t'=1}^t \bar{\alpha}_{t'}$ . Then  $x_t$  and  $e_t$  can be obtained with  $x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon$  and  $e_t = \sqrt{\bar{\alpha}_t}e_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon$ , where  $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ .

**3.3.2 Reverse Process.** The reverse process aims to iteratively denoise  $x_t$  and  $e_t$  over  $t$  steps, ultimately recovering the original representations  $x_0$  and  $e_0$ . Instead of treating node and edge independently like standard diffusion model, we model the node and edge generation as a graph-conditioned denoising diffusion process, leveraging graph structure to capture dependencies and enable context-aware modeling. In specific, for the generation of nodes, we first obtain the graph context representation on the known graph  $G_{va} = (\{V^v, V^a\}, E^{va})$ . The visual nodes and anchors are jointly encoded via GCN networks to obtain the latent node representations to form the condition vector:

$$c_{va} = \text{GCN}([V^v; V^a], G_{va}), \quad (5)$$

which captures the structural and semantic context of the known graph and is used as a condition context for the node reverse process. With the generated textual semantic nodes, we compute the similarity score with anchors:

$$s_{ta} = \text{sim}(V^t, V^a), \quad (6)$$

which reflects how the generated nodes related to anchors and serves as the condition context for the edge diffusion process. This condition mechanism allows the model to generate nodes and edges that are semantic grounded and structure coherent. We incorporate these graph condition context information as additional conditions to explicitly guide the conditional generation of node and edge representations via a tailored conditional estimator. The conditional generation process at each step  $t$  can be formulated as:

$$\begin{cases} p_\theta(x_{t-1}|x_t, c_{va}) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, c_{va}, t), \Sigma_\theta(x_t, t)) \\ p_\theta(e_{t-1}|e_t, s_{ta}) = \mathcal{N}(e_{t-1}; \mu_\theta(e_t, s_{ta}, t), \Sigma_\theta(e_t, t)) \end{cases} \quad (7)$$

where  $\mu_\theta$  and  $\Sigma_\theta$  denote the mean and covariance values of the predicted Gaussian distribution, respectively. Similar to the DDPM framework, we adopt the MSE loss to measure the diffusion process, which is formulated as:

$$\begin{aligned} \mathcal{L}_{diff} = & \mathbb{E}_{x_0, \epsilon, t} [\|f_{\theta_n}(x_t, c_{va}, t) - x_0\|^2] \\ & + \mathbb{E}_{e_0, \epsilon, t} [\|f_{\theta_e}(e_t, s_{ta}, t) - e_0\|^2], \end{aligned} \quad (8)$$

where  $f_{\theta_n}$  and  $f_{\theta_e}$  are the conditioned denoisers. Here we utilize MLP as  $f_{\theta_n}$  for node denoising to efficiently fuse visual and anchor-based semantic information at the node level, where structural dependencies are weak. In contrast, we adopt GCN as  $f_{\theta_e}$  for edge denoising to explicitly model relational dependencies and preserve graph structural consistency during diffusion.

**3.3.3 Soft Pruning Mechanism.** The above diffusion module generates the missing nodes and edges and forms the complete multimodal graph  $G$ . To evaluate how meaningful and valid each generated target node is, we further set a validity prediction module to assess the validity of each generated node and performs soft pruning to filter out noisy elements. The valid module predicts a

valid score for each generated node. To perform differentiable and controllable pruning, we adopt the Gumbel-Sigmoid with Top- $k$  selection:

$$\tilde{p}_i = \sigma\left(\frac{MLP(\mathbf{h}_i) + g_i}{\tau}\right), \quad g_i \sim \text{Gumbel}(0, 1), \quad (9)$$

where MLP is utilized to predict the valid score.  $k = \max(1, \lfloor r \cdot n_t \rfloor)$ , where  $r$  is the pruning ratio. We select the top- $k$  nodes based on  $\tilde{p}_i$  and define a binary mask accordingly to achieve pruning and reduce the complexity of the graph:

$$m_i = \begin{cases} 1 & \text{if } i \in \text{Top-}k(\tilde{p}) \\ 0 & \text{otherwise} \end{cases}. \quad (10)$$

The valid module serves as a post-diffusion filter that selects semantically meaningful target nodes based on the confidence scores, thus further refining the quality of the generated graph.

After filtering out low-quality nodes through the pruning mechanism, we obtain the final refined multimodal graph  $G$ . To ensure that the final representation captures the global graph context and the task-specific semantic, we apply a GCN to refine the representation of the generated  $V^t$  based on the full graph structure:

$$\mathbf{h}_i = \text{GCN}(V_i^t, \{V_i^{v,a}, i \in \mathcal{N}(i)\}), \quad (11)$$

where  $\mathcal{N}(i)$  denotes the neighbors of  $V_i^t$  in the graph  $G$ . This ensures that the generated visual counterpart representation is contextually encoded on the basis of the whole graph structure that is more adaptive to the caption task.

### 3.4 Caption Generation

For caption generation, we utilize a Bert [6]-like structure as the caption generator, which takes the text embedding of a partially masked sentence  $\mathbf{v}_{mask}$ , the visual representation  $\mathbf{V}$ , and the obtained visual counterpart representation  $\mathbf{h}$  as input and output the final generated caption. In specific, we mask a portion of words by replacing them with the special token  $[MASK]$ , the caption generator then predicts the true values of these masked words to reconstruct the full caption. We train our model through masked language modeling and optimizing the captioning loss  $\mathcal{L}_{cap}$ :

$$\mathcal{L}_{cap} = f_{gen}(\mathbf{v}_{mask}, \mathbf{V}, \mathbf{h}). \quad (12)$$

We utilize cross-entropy to optimize the captioning loss, which can be further denoted as:

$$\mathcal{L}_{cap} = - \sum_{i=1}^L \log(P(w_i | w_{<i}, \mathbf{V}, \mathbf{h})), \quad (13)$$

where  $L$  is the length of the caption, and  $w_{1:L}$  is the target ground-truth caption. Then the final loss is defined as:

$$\mathcal{L} = \mathcal{L}_{ali} + \mathcal{L}_{dif} + \mathcal{L}_{cap}. \quad (14)$$

## 4 Experiments

### 4.1 Datasets and Metrics

The proposed model is compared with baselines on widely used video captioning datasets. These datasets provide diverse video content and corresponding textual descriptions, allowing for a comprehensive evaluation of the method's performance in generating accurate and contextually relevant captions. In specific:

**Microsoft Research Video Description Corpus (MSVD)** dataset [3] comprises 1,970 open-domain video clips sourced from YouTube. On average, each video lasts between 10 to 25 seconds and primarily features a single activity. Each video clip is accompanied by approximately 40 ground-truth captions. We adhere to the standard data split used in previous works, which includes 1,200 training videos, 100 validation videos, and 670 test videos.

**Microsoft Research Video to Text (MSR-VTT)** dataset [49] contains 10,000 open-domain video clips, spanning a variety of content across 20 categories. On average, each video lasts about 20 seconds and is accompanied by approximately 20 ground-truth captions. We utilize the standard data split established in previous works, which includes 6,513 training videos, 497 validation videos, and 2,990 test videos.

**VATEX** dataset [45] is a large-scale multilingual video description dataset, which contains over 41,250 open-domain videos. Each video clip has 20 ground-truth captions in both English and Chinese. We only use the English corpus and utilize the standard data split established in previous works, which includes 25,991 training videos, 3,000 validation videos, and 6,000 public test videos.

**Basketball Highlight Commentary (BH-Commentary)** dataset [56] consists of 4,396 NBA highlight video clips from online video platform, containing six types of basketball techniques. On average, each video lasts about 10.5 seconds and is annotated by the professional commentators. We utilize the standard data split established in previous work, which includes 2,860 training videos, 216 validation videos, and 1,320 test videos.

In addition, several widely used evaluation metrics are used to thoroughly evaluate the performance of our proposed model, including BLEU@4 [38], METEOR [2], Rouge-L [26], and CIDEr [43]. To ensure a fair comparison, we utilize the publicly available code provided by the Microsoft COCO Caption Evaluation server<sup>1</sup> to compute all these metrics. All results are reported in %, and '-' indicates that the models do not adopt the corresponding metric.

### 4.2 Implementation Details

Our model is implemented in PyTorch [39], and we use deepspeed for automatic mixing precision training. The visual encoder is initialized with Kinetics-600 [17] pre-trained weights, and textual encoder is initialized from pre-trained BERT-Base [6]. The caption generator adopts a transformer-based structure with 768 dimensional hidden states. In addition, we adopt AdamW [18] optimizer and use a learning rate warm-up during the early 10% training steps followed by linear decay.

During training, the model takes visual and textual inputs. We utilize a causal self-attention mask to ensure a unidirectional generation process, i.e., each caption token can only attend to previously generated tokens. Additionally, all textual embedding is given full attention to visual representation, allowing the model to effectively integrate both visual and textual information to generate accurate and informative captions. During inference, the model only takes visual input, constructing the multimodal graph on the basis of visual input and the learned anchors, and generating the caption based on the final generated multimodal graph. The generator produces

<sup>1</sup><https://github.com/tylin/coco-caption>

**Table 1: Comparison of our proposed model with baselines on MSVD and MSR-VTT datasets.**

| Methods      | MSVD        |             |             |              |             | MSR-VTT     |             |             |             |             |
|--------------|-------------|-------------|-------------|--------------|-------------|-------------|-------------|-------------|-------------|-------------|
|              | BLEU@4      | METEOR      | Rouge-L     | CIDEr        | Avg.        | BLEU@4      | METEOR      | Rouge-L     | CIDEr       | Avg.        |
| OSTG [57]    | 57.5        | 36.8        | -           | 92.1         | 62.1        | 41.9        | 28.6        | -           | 48.2        | 39.6        |
| STG-KD [37]  | 52.2        | 36.9        | 73.9        | 93.0         | 64.0        | 40.5        | 28.3        | 60.9        | 47.1        | 44.2        |
| ORG-TRL [59] | 54.3        | 36.4        | 73.9        | 95.2         | 65.0        | 43.6        | 28.8        | 62.1        | 50.9        | 46.4        |
| SGN[41]      | 52.8        | 35.5        | 72.9        | 94.3         | 63.9        | 40.8        | 28.3        | 60.8        | 49.5        | 44.9        |
| MGRMP[5]     | 55.8        | 36.9        | 74.5        | 98.5         | 66.4        | 41.7        | 28.9        | 62.1        | 51.4        | 46.0        |
| M2DT[60]     | 52.0        | 36.5        | 72.8        | 92.6         | 63.5        | 42.7        | 29.4        | 61.9        | 50.1        | 46.0        |
| LSRT [25]    | 55.6        | 37.1        | 73.5        | 98.5         | 66.2        | 42.6        | 28.3        | 61.0        | 49.5        | 45.4        |
| TKVC [47]    | 50.5        | 34.5        | 71.7        | 84.3         | 60.3        | 43.0        | 28.7        | 62.2        | 51.8        | 46.4        |
| HMN [53]     | 59.2        | 37.7        | 75.1        | 104.0        | 69.0        | 43.5        | 29.0        | <b>62.7</b> | 51.5        | 46.7        |
| BME-WCO [30] | 51.5        | 35.1        | 72.5        | 100.2        | 64.8        | 40.6        | 28.1        | 61.2        | 53.4        | 45.8        |
| RSFD [62]    | 51.2        | 35.7        | 72.9        | 96.7         | 64.1        | 43.4        | 29.3        | 62.3        | 53.1        | 47.0        |
| TextKG [10]  | 60.8        | 38.5        | 75.1        | 105.2        | 69.9        | 43.7        | 29.6        | 62.4        | 52.4        | 47.0        |
| CAT [10]     | 59.9        | 41.7        | 78.4        | 122.9        | 75.7        | 42.1        | <b>30.2</b> | 62.5        | 54.5        | 47.3        |
| GSEN [34]    | 58.8        | 37.6        | 75.2        | 102.5        | 68.5        | 42.9        | 28.4        | 61.7        | 51.0        | 46.0        |
| MAN [16]     | 59.7        | 37.3        | 74.3        | 101.5        | 68.2        | 41.3        | 28.0        | 61.4        | 49.8        | 45.1        |
| ASGNet [31]  | 55.2        | 36.6        | 74.3        | 101.8        | 67.0        | 43.4        | 29.6        | 62.5        | 52.6        | 47.0        |
| VCRN [54]    | 59.1        | 37.4        | 74.6        | 100.8        | 68.0        | 41.5        | 28.1        | 61.2        | 50.2        | 45.3        |
| MGCDVc       | <b>61.6</b> | <b>42.2</b> | <b>78.9</b> | <b>124.3</b> | <b>76.8</b> | <b>43.8</b> | 29.9        | <b>62.7</b> | <b>54.8</b> | <b>47.8</b> |

one text token at a time and continues until the model outputs the ending token *[EOS]* or reaches the maximum caption length.

**Table 2: Comparison of our proposed model with baselines on VATEX dataset.**

| Model            | BLEU@4      | METEOR      | Rouge-L     | CIDEr       | Avg.        |
|------------------|-------------|-------------|-------------|-------------|-------------|
| VATEX [45]       | 28.4        | 21.7        | 47.0        | 45.1        | 35.6        |
| ORG-TRL [59]     | 32.1        | 22.2        | 48.9        | 49.7        | 38.2        |
| Support-set [40] | 32.8        | 22.2        | 49.1        | 51.2        | 38.8        |
| OpenBook [58]    | 33.9        | 23.7        | 50.2        | 57.5        | 41.3        |
| HRNAT [8]        | 32.5        | 22.3        | 49.0        | 50.7        | 38.6        |
| HMN [53]         | 33.2        | 22.7        | 49.3        | 51.6        | 39.2        |
| RLHMN [24]       | 35.3        | 23.1        | 50.9        | 58.8        | 42.0        |
| MAN [16]         | 32.7        | 22.4        | 49.1        | 48.9        | 38.3        |
| VCRN [54]        | 32.4        | 22.4        | 48.9        | 49.9        | 38.4        |
| MGCDVc           | <b>36.6</b> | <b>25.3</b> | <b>52.1</b> | <b>68.4</b> | <b>45.6</b> |

### 4.3 Main Results

Several latest and representative baseline models are selected to evaluate the effectiveness of the proposed model. Table 1 and Table 2 display the result of the performance comparison between our model and the representative baselines on MSVD, MSR-VTT and VATEX datasets, respectively. From the tables, we can observe that our proposed model achieves better performance on the two datasets compared to the selected baselines. And we also present the average scores on all evaluation metrics in the last column of the table under the two datasets. As shown, our model consistently achieves the best average performance, which demonstrates that

our model achieves better performance than the selected baselines, confirming its superiority in generating accurate and semantically rich video captions. Moreover, we compare the proposed model with several baselines on BH-Commentary dataset, which focuses more on the interaction between objects contained in the video. The comparison results, as listed in Table 3, demonstrate the superior performance of our model compared to existing work. According to the tables, it is not difficult to find that our model demonstrates performance that is superior to or comparable to the selected baseline models. The reason why our model achieves better performance compared to baselines can be summarized as follows: (1) The construction of multimodal graph based on the learned anchors enables multimodal information to be interactively fused and modeled, laying the foundation for generating high-quality captions, compared with single visual/textual graph representation learning in previous work. (2) Graph conditioned diffusion strategy captures the structural semantic correlation contained in multimodal contents, enabling the generation of textual semantic nodes and providing semantic guidance for caption generation. (3) The pruning mechanism filters out low-quality nodes, refining the generated graph structure and promoting meaningful semantic focus for caption generation.

### 4.4 Ablation study

**4.4.1 Model components.** To investigate the effectiveness of each component of the proposed model, we conduct experiments with different model settings by removing each component of the proposed model. The settings include "base", which removes all the components that utilize the extracted visual feature for caption generation; "w/o anchor", which means removing the anchors and

**Table 3: Comparison of our proposed model with baselines on BH-Commentary dataset.**

| Model            | BLEU@4      | METEOR       | Rouge-L      | CIDEr        | Avg.         |
|------------------|-------------|--------------|--------------|--------------|--------------|
| RecNet [44]      | 2.02        | 8.98         | 27.62        | 6.37         | 11.25        |
| NACF [52]        | 2.00        | 11.64        | <b>31.20</b> | 7.93         | 13.20        |
| SGN [41]         | 3.41        | 10.85        | 27.98        | 11.36        | 13.40        |
| UniVL+MELTR [19] | 1.09        | 11.12        | 24.54        | 10.75        | 11.88        |
| Swinbert [27]    | 3.82        | 12.73        | 28.49        | 11.12        | 14.04        |
| RSFD [62]        | 3.91        | 12.40        | 30.10        | 11.47        | 14.47        |
| BCG [56]         | 4.13        | 12.89        | 28.69        | 12.19        | 14.48        |
| MGCDVc           | <b>4.34</b> | <b>12.90</b> | 28.64        | <b>12.58</b> | <b>14.62</b> |

only using the visual input as the condition for node diffusion generation; "w/o diffusion", which means removing the diffusion process that utilize the visual input and learned anchors for caption generation; "w/o pruning", which means removing the soft pruning mechanism that construct the final multimodal graph without filtering the low quality generated textual semantic nodes. The results are listed in Table 4, which clearly illustrates the positive impact of the designed multimodal graph conditioned diffusion on model performance. Specifically, compared with the base setting, the whole model achieves significant performance improvements, demonstrating the effectiveness of our multimodal graph conditioned diffusion model in the caption generation task. When removing the learned anchors, the performance drops, which indicates that the latent concept anchors play a critical role in bridging the visual and textual modalities to enable more effective multimodal reasoning. When removing the diffusion process, the performance of our model is affected, showing that the conditioned diffusion process is essential for modeling the complex semantic structure underlying the cross-modal content. When removing the pruning mechanism, our model's performance has been limited, which suggests that filtering out noisy or irrelevant nodes helps construct a more informative and compact multimodal graph.

**Table 4: Ablation study on the components of the proposed model on MSVD dataset.**

| setting       | BLEU@4 | METEOR | Rouge-L | CIDEr |
|---------------|--------|--------|---------|-------|
| base          | 58.1   | 40.3   | 77.3    | 116.2 |
| w/o anchor    | 59.9   | 40.9   | 77.8    | 121.1 |
| w/o diffusion | 61.0   | 41.2   | 78.1    | 120.8 |
| w/o pruning   | 61.5   | 41.7   | 78.0    | 119.1 |
| whole model   | 61.6   | 42.2   | 78.9    | 124.3 |

Moreover, we test the model with the setting of unconditioned diffusion and unweighted graph, the results are shown in Table 5, where we remove the condition during diffusion and set all the existing edge weights of the graph as 1, respectively. In the unconditioned diffusion setting, we remove the condition during the diffusion process, which leads to a drop in model performance, highlighting the importance of condition guidance in generating semantic relevant nodes and measuring the weights of edges. In the

unweighted graph setting, we set all existing edge weights in the multimodal graph to 1, which would ignore the learned modality-aware relations between nodes. This also results in degraded performance, demonstrating that the weighted edge structure contributes significantly to capturing the relevance of cross-modal interactions.

**Table 5: Ablation study on the different model component setting on MSVD dataset.**

| setting          | BLEU@4 | METEOR | Rouge-L | CIDEr |
|------------------|--------|--------|---------|-------|
| w/o condition    | 59.6   | 41.7   | 78.2    | 120.5 |
| unweighted graph | 58.7   | 41.2   | 77.8    | 118.9 |
| whole model      | 61.6   | 42.2   | 78.9    | 124.3 |

**Table 6: Ablation study on the number of anchors on MSVD dataset.**

| num of $n_a$ | BLEU@4 | METEOR | Rouge-L | CIDEr |
|--------------|--------|--------|---------|-------|
| 10           | 59.9   | 41.3   | 77.9    | 119.8 |
| 20           | 61.6   | 42.2   | 78.9    | 124.3 |
| 30           | 60.3   | 41.8   | 78.4    | 121.2 |
| 40           | 59.7   | 41.2   | 78.1    | 120.3 |
| 50           | 59.5   | 40.8   | 77.6    | 116.5 |

**Table 7: Ablation study on the number of diffusion steps on MSVD dataset.**

| num of step | BLEU@4 | METEOR | Rouge-L | CIDEr |
|-------------|--------|--------|---------|-------|
| 10          | 58.1   | 40.9   | 77.8    | 117.4 |
| 20          | 59.4   | 41.0   | 78.0    | 120.8 |
| 30          | 61.6   | 42.2   | 78.9    | 124.3 |
| 40          | 60.4   | 41.0   | 78.1    | 116.6 |
| 50          | 58.2   | 40.7   | 77.6    | 118.7 |

**4.4.2 Number of anchors.** To investigate the effect of the number of anchors on the performance of the model, we conduct experiments with different values of  $n_a \in [10, 50]$  to explore how the variation of the number of anchors influences the results. The results are shown in Table 6, as the number of anchors increases from 10 to 20, the model performance improves, indicating that a greater number of latent anchors enhances the model's ability to capture diverse semantic concepts and effectively bridge the visual and textual modalities. However, when  $n_a$  exceeds 20, we observe a gradual decline in performance. This suggests that introducing too many anchors may introduce noise or redundancy, which would hinder the model to focus on the most informative semantic concepts. Therefore, we conclude that  $n_a = 20$  provides an effective balance between expressiveness and efficiency, and is sufficient for modeling the cross-modal content in our framework.

**4.4.3 Number of diffusion steps.** To investigate the effect of diffusion steps on model performance, we perform experiments with different values of diffusion steps in [10, 50] to explore how the variation in the diffusion steps influences the results. The results are shown in Table 7, the model performance improves as the number of diffusion steps increases from 10 to 30, indicating that more steps allow for a more refined and stable generation process. However, beyond 30 steps, the performance begins to decline, which is possibly due to over-smoothing or the accumulation of noise. These findings suggest that 30 diffusion steps provide an optimal balance between generation quality and computational efficiency, enabling the model to form more accurate and informative textual semantic nodes and their relations.



Figure 3: Qualitative results of the proposed model. The first line shows the video frames, the second line shows the ground-truth (GT), and the last two lines are the results of the baseline and our model, respectively.

## 4.5 Qualitative Analysis

Figure 3 exhibits several qualitative examples generated by our proposed model and the baseline. We can observe that our model has the ability to identify visual elements in videos and accurately describe them. In specific, as shown in case (a), the ground-truth captions show the activity of playing between a baby and a dog. The baseline model, however, only mentions the dog and misses the key interaction with the baby. In contrast, our model outputs the caption “a baby is playing with a dog,” which exactly consists with the ground-truth. This shows that our model not only identifies both main objects but also captures their relationship accurately,

demonstrating clear improvement in semantic completeness and alignment with the ground-truth compared to the baseline. For case (b), the ground-truth captions describe a man both singing and playing the guitar, with equal emphasis on the two actions. The baseline model misidentifies the object and lacks the singing action, outputting the caption that is not entirely correct. In contrast, our model generates “a man is singing and playing the guitar,” which is semantically accurate and nearly identical to one of the ground-truth captions. This shows that our model captures both the correct actions and the correct object, achieving strong alignment with the ground-truth and a better performance over the baseline. Case (c) involves a basketball play segment, the ground-truth caption provides a rich sequence of actions involving multiple defenders and a finishing slam dunk. The baseline focuses mainly on the dunk itself, neglecting the crucial defensive confrontation and the offensive pressure contained in the highlight moment. In contrast, our model not only depicts the decisive dunk but also emphasizes the direct clash with the defender, which better reflects the competitive dynamics of the segment. Although less detailed than ground-truth, our model captures the essence of the offense–defense interaction, showing the advantage in generating semantically richer and more contextually aligned descriptions.



Figure 4: Failure case of the proposed model.

**Failure Case.** Figure 4 shows a failure case of our proposed model. Compared with the GT, the generated caption is relatively coarse and omits some fine-grained visual and semantic details, such as the specific scene context and descriptive attributes. This suggests that the model may have room for improvement in capturing detailed visual information and richer semantics.

## 5 Conclusion

In this paper, we present a multimodal graph conditioned diffusion model for video captioning, where we incorporate graph-based relational reasoning with diffusion-based generative modeling to jointly model cross-modal relationships and capture latent semantic structure for caption generation. In the future, we will focus on exploring the powerful semantic understanding capacity of large visual language models in modeling multimodal representation in the diffusion process, enabling more accurate caption results.

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